



## CONTACT

### EMAIL

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### WEB

hsandmann.github.io

### GITHUB

hsandmann

### LINKEDIN

in/hsandmann

### PROJECT

graphole.com

## SKILLS

### Languages

Java Python Go Kotlin C

C++ JavaScript SQL

### Machine Learning & AI

Deep Learning Neural Networks

Computer Vision NLP PCA SVM

Clustering

### Backend & Architecture

Spring Boot Spring Cloud

Microservices Design Patterns

Software Architecture

### Data & Infrastructure

PostgreSQL Redis Kafka

RabbitMQ Docker Kubernetes

Spark MongoDB Neo4j

### Domains

Banking & Fintech

Cryptography (PKI/HSM)

High-Performance Computing

POS/ERP

## LANGUAGES

Portuguese Native

English Advanced

German B1

# Humberto Sandmann

## Computer Scientist & Professor

## PROFILE

Senior software architect and machine-learning engineer with 20+ years building high-stakes systems across banking, fintech, and retail — core banking, cryptography (PKI/HSM), high-performance computing, computer vision, and cloud-native microservices. D.Sc. in neural computing; also a university professor teaching software design, ML, and distributed systems.

## EXPERIENCE

### Founder · H-Sandmann

2006–present

Independent consultancy in machine learning and software architecture.

Machine Learning

Software Architecture

### Co-Founder · SelsanTech

2012–2023

Co-founded a venture delivering machine-learning and software-architecture solutions.

Machine Learning

Software Architecture

### Senior Software Architect & Machine Learning Engineer · RockSpoon

2018–2019

Architected POS/ERP/payroll platforms and computer-vision/ML features for a restaurant-tech company.

POS

ERP

JavaEE

PostgreSQL

OpenCV

Computer Vision

Go

Android

Kotlin

### Senior Consultant, Computational Optimization · Banco Votorantim

2015–2016

Return-on-equity optimization on high-performance computing infrastructure for a major bank.

RoE

High-Performance Computing

J2EE

Spring Boot

Jenkins

Test Automation

### Senior Consultant, Computational Optimization · TecBan

2011–2011

ATM network optimization with in-memory databases and high-performance computing.

ATM

J2EE

In-Memory DB

High-Performance Computing

### Senior Software Engineer / Architect · ABN Amro Bank

2006–2007

Core-banking, stock-market and trading systems with cryptography and PKI.

Core Banking

Trading

Cryptography

HSM

BouncyCastle

PKI

PKCS

Java

Servlet

### Senior System Analyst · Politec

2005–2006

### Senior Software Engineer / Architect · EverSystems

2004–2005

### Senior System Analyst · Central Online

2002–2003

### Senior System Analyst · Datamace

1999–2002

### Computer Programmer · Heraeus Electro-Nite

1998–1999

## EDUCATION

**D.Sc.** · Electrical Engineering — Polytechnic School, University of São Paulo

2007–2012

**M.Sc.** · Electrical Engineering — Polytechnic School, University of São Paulo

2003–2006

**B.Sc.** · Computer Science — Faculdade de Engenharia Industrial (FEI)

1999–2002

Full academic CV, publications and teaching record at [hsandmann.github.io](https://hsandmann.github.io)